

Helix Constitution – Universal GEMINI.md

Field	Value
Revision	
Created	- -
Last modified	- - T : : Z
Status	active

Status summary

Added § . . . – Device-independent host-side rendered-UI visual-proof mandate (User mandate – –): every change to ANY user-facing UI surface MUST be proven by DEVICE-INDEPENDENT host-side RENDERED PIXELS before claimed correct – the real component rendered to a PNG ON THE HOST (no device/emulator/running app; Compose→Roborazzi/Paparazzi, web→Playwright/Storybook, SwiftUI→snapshot-testing, equiv per stack) for EVERY screen×state×{light,dark} theme, dual-validated by (i) golden image-diff AND (ii) an OCR/vision oracle reading rendered text+labels+control bounds (NO overlap / label-over-label / clipping / off-screen / collapsed-or-giant-unbounded widget). VALUE/token-equality / property-assertion UI tests are FORBIDDEN as the PROOF a UI is correct (forensic FACT – – : hex/sp/dp value-equality unit tests stayed GREEN while the operator opened a broken giant-button screen) – may supplement, NEVER substitute the rendered-pixel proof; self-validated golden-good/golden-bad analyzer §" " . . . (); device-offline is NEVER a valid skip (host-render IS the device-independent path); COMPLEMENTS not replaces § . . . /. /. /. live-device¹⁰⁴ recording + § . . . OpenDesign tokens + § . . . exported-doc visual validation. Propagation gate CM-COVENANT– – –PROPAGATION +

recommended gate CM-HOST-RENDERED-UI-VISUAL-PROOF" + paired \$. mutation.
Classification: universal
(\$. .). Prior round:
\$. . REPEALED (operator
decision - -):
Universal Semgrep static-analysis
mandate repealed – Semgrep no longer
mandatory; scaffolding, submodules/
semgrep, MCP/pre-commit/PATH wiring,
docs_chain semgrep_status context, and
CM-COVENANT- - -PROPAGATION/
CM-SEMGREP-WIRED gates removed. Added
\$. . mirror – Universal
Media Validation: every recorded
artifact MUST pass MEDIA VALIDATION
pipeline (OCR/transcription/text parsing
vs patterns, self-validated analyzer,
structured verdict, pinpoint on FAIL,
real-time trigger, \$. mutation).
Added \$. . mirror –
Universal Auto-Propagation Hook:
post_update_hook.sh detects/registers/
installs skills/MCP/hooks/scripts;
consumer invokes after every pull. Added
\$. . mirror – Universal
Independent Verification Agent: every
output¹¹ passes INDEPENDENT verifier
(\$. . /.), zero-finding G0.
All Classification: universal
(\$¹¹ . .). Prior round: Added
\$. . - mirror entries
– Vision-verified recording + HelixQA
bridge (User mandate
- -): every video
recording MUST be processed through a
vision/OCR pipeline reading on-screen
content against SPECIFY-phase expected
patterns; ≤ s frame capture, self-

validated analyzer, per-frame PASS/FAIL with evidence path. Rootless container runtime (User mandate - -): Podman rootless mandatory; rootful Docker/sudo FORBIDDEN unless § . . ; vasic-digital/containers sole orchestration layer. OpenDesign UI design system (User mandate - -): mandatory UI design-and-refinement system for all user-facing interfaces; design tokens for color/typography/spacing; light+dark themes; extend upstream per § . . . All three

Classification: universal (§ . .). GEMINI.md mirrors updated in lockstep per § . . . Prior round: New Gemini CLI carrier of the Helix Constitution – added by § . . (universal action-prefix system) so Gemini CLI gets the LAYER- action-prefix recognition floor out of the box, plus the § . . token-efficiency mandate mirror. This file is one of the four canonical agent context carriers (CLAUDE.md , AGENTS.md , QWEN.md , GEMINI.md) the constitution maintains per § . . ; the textual identity of the action-prefix recognition block across all four IS the LAYER- universality. This GEMINI.md is a thin carrier – the full universal rule set is in Constitution.md (canonical) and the sibling carriers; Gemini CLI reads this file as its context carrier. Back-filled § . . - to lockstep (§ . .).

Field	Value
Issues	none
Issues summary	—
Fixed	\$. . + \$. . mirrors
Fixed summary	Carrier created in lockstep with the Constitution.md \$. . + \$. . additions.
Continuation	—

How inheritance works

This GEMINI.md is the **Gemini CLI carrier** of the Helix Constitution submodule. It is one of the four canonical agent context carriers (constitution/CLAUDE.md , constitution/AGENTS.md , constitution/QWEN.md , constitution/GEMINI.md) – these ARE the canonical root per \$. . , NOT consumer extensions. The full universal rule set lives in constitution/Constitution.md ; the sibling carriers (CLAUDE.md / AGENTS.md / QWEN.md) carry the same content for their respective agents. A consuming project’s own repo-root GEMINI.md (if any) opens with the inheritance pointer and adds only project-specific rules; THIS file is the canonical carrier and does not.

For agents driven from a consuming project, the project’s repo-root carrier SHOULD start with a clearly-marked inheritance pointer:

```
## INHERITED FROM constitution/GEMINI.md
```

```
All rules in `constitution/GEMINI.md` (and the `constitution/  
Constitution.md`  
it references) apply unconditionally. The project-specific rules below  
extend them.
```

```
1 1 4 1 4 0
```

```
1
```

Action-prefix recognition (§ . . – LAYER-)

The block below is the **canonical LAYER- action-prefix recognition instruction**, embedded VERBATIM (identical text) into every agent context carrier the constitution maintains. That textual identity across all four carriers IS the LAYER- universality: Gemini CLI reads this block via this carrier and applies the prefix expansion itself, so the action-prefix system works on Gemini CLI with zero extra setup.

§ . . – Universal action-prefix system

(ACTION_NAME ::) (User mandate, – – ;

GRAMMAR_ADDENDUM – – ; arrow form

– –). When a user prompt's FIRST non-blank line starts with a recognised action prefix, you MUST: () look the action token up in the action registry constitution/actions/registry.yaml (or \$HELIX_ACTION_REGISTRY); () if it is a registered action, REPLACE the prefix with that action's expansion text and apply its rules ; () execute the remainder of the prompt under the expanded instruction. **Five EQUIVALENT forms** – same action, same expansion, same execution: () ACTION_NAME :: <rest> (bare ::), () PREFIX::ACTION_NAME :: <rest> (namespaced ::), () / ACTION_NAME <rest> (bare slash), () /PREFIX::ACTION_NAME <rest> (namespaced slash), () ACTION_NAME ---> <rest> (bare arrow) ≡ PREFIX::ACTION_NAME ---> <rest> (namespaced arrow). Thus BACKGROUND :: x ≡ DEFAULT::BACKGROUND :: x ≡ /BACKGROUND x ≡ /DEFAULT::BACKGROUND x ≡ BACKGROUND ---> x ≡ DEFAULT::BACKGROUND ---> x . PREFIX is an action NAMESPACE; the reserved default namespace is **DEFAULT** , and an action runs WITH or WITHOUT the prefix. Grammar (all hold): anchored at the FIRST non-blank line only (mid-prose tokens never match); the action token AND the namespace are UPPERCASE-only [A-Z][A-Z0-9_]* (lowercase never matches); the namespace separator :: inside the token carries NO surrounding spaces (PREFIX::ACTION_NAME), DISTINCT from the action-body separator " :: " (one ASCII space on each side of :: – avoids C++ Foo::Bar , YAML key: value , URLs) in forms / , the arrow-body separator " ---> " (one ASCII space on each side) in form , and the slash-body separator (one space) in forms / ; stacked prefixes (A :: B :: rest) apply outer-to-inner, left-to-right (expand A , re-scan, expand B , then the residual is the task); a leading \ escapes

the prefix for the `::`, arrow, and slash forms
`( \BACKGROUND :: x , \BACKGROUND ---> x , \/BACKGROUND x -`
 treat literally, strip the backslash, NO expansion) so
 action names can be discussed. **Conflict rule (slash**
form): `/ACTION_NAME` (form) is honored as the action
 ONLY when `ACTION_NAME` (case-folded) does not collide with
 a built-in/host slash command (registry `slash_bare: auto`
`+ slash_conflicts: [..]`); form `( /PREFIX::ACTION_NAME )`
 is ALWAYS unambiguous and always honored. An unknown
 token that matches the grammar shape (any of the
 forms) but is NOT registered is NEVER silently expanded
 or silently dropped – ask which registered action was
 meant (`$ . . / $ . .`) or treat it as
 a literal prompt, NEVER invent an expansion
`( $ . . )`; any prompt not satisfying the grammar is
 an ordinary prompt and the system is a no-op. The
 registered action **BACKGROUND** expands to: *“The following*
prompt that we will provide MUST BE executed in
background in parallel with all main work streams using
the subagents-driven development approach! All work done
MUST PRODUCE rock solid evidence covered with hard
physical proof(s) that all done is working as expected
and as specified without any false results and without
any bluff!” (composes `$ . . . / $ . .`
`subagent-driven, $ . . / $ . .`
`parallel streams, $ . . background execution,`
`$ . . / $ . . / $ . .`
`captured physical evidence, $ . anti-bluff`). A
 second registered action **REMINDER** re-surfaces
 previously-scheduled, CRITICAL, status-UNCERTAIN work:
 FIRST verify the ACTUAL current status from captured
 evidence (never assume done or not-done – `$ . .`
 no-guessing), THEN act on the delta – report the captured
 proof if genuinely complete, resume from the exact point
 if partial, action it NOW if not started, or surface the
 block per `$ . . / $ . .` – always

11 4 87 11 4 94 11 4 97
11 4 100 11 4 106
11 4 130 11 4 147

producing a status verdict (composes \$. . /
\$. . / \$ 11 4 35 . / \$. . /
\$. . / \$. . /
\$ 11 4 20 . / \$. .). The system is
UNIVERSAL (every CLI agent reads this block via its
context carrier per \$. .), extensible (new
action = new registry row), decoupled + reusable
(\$. .), and loads out-of-the-box.
Classification: universal (\$. .). **Canonical
authority:** constitution submodule Constitution.md
\$. . . Non-compliance is a release blocker. No
escape hatch – no --skip-action-prefix , --ignore-prefix ,
--no-registry , --invent-expansion-OK , --single-layer-only
flag.

11 4 141

Token-efficiency mandate (\$. .)

\$. . – Token-efficiency mandate (research-derived
+ operator mandate, – –). Every project
worked on by AI coding agents MUST cut token spend (input AND
output) toward – % of current (a – %
reduction) WITHOUT degrading quality/performance/safety or
breaking any existing mechanism, via a composable, safety-
ranked measure set: () **prompt-cache the static governance
prefix** – the always-loaded governance forms a byte-stable
cache-breakpointed prefix with no volatile bytes ahead of it;
cache reads cost ~ . × base input (the dominant cost driver
– measured ~ K tokens of governance re-sent every turn,
externally corroborated by Claude Code issue);
caching is transparent so it removes no rule, weakens no
gate, changes no verdict – only billing (PRIMARY, biggest +
safest lever); the operative discipline is to **batch
governance edits and keep CLAUDE.md + the constitution byte-
stable mid-session** – every governance edit busts the prompt

cache (BASELINE.md finding); () **subagent model-tiering + output-to-file** – mechanical non-judgment work (search/grep/status/doc-export/read-only probes) to a Haiku-class model, the strong model RESERVED for all reasoning/verdicts/fix-design (\$. . .)/code-review (\$. . .)/demotion (\$. . .), large output persisted to a file not an inline – K-token transcript; the cheap model never emits a PASS so \$. . . + anti-bluff are untouched; () **thin always-loaded INDEX + on-demand detail** – concise index (one line per fix/anchor, EACH carrying the literal 11.4.N token so propagation gates pass) with the canonical full text kept gate-scanned in constitution/Constitution.md and reachable in one hop – a de-duplication realising \$. . . , never a deletion; () **CodeGraph/retrieval-first over full-file loading** (\$. . . / \$. . . / \$. . .); () **output-token reduction** – terse status + effort:"low" on the mechanical allowlist only; () **tool-call batching + no re-reads**; () **compaction/context-editing for long sessions**. **Mandatory measured proof:** a token-accounting harness measures tokens-per-development-cycle BEFORE vs AFTER on a frozen deterministic workload from the authoritative usage object (input/cache_read/cache_creation/output split; NEVER tiktoken, NEVER the client-side cost estimate), reproduced N times (\$. . .), pass = AFTER ≤ % of BEFORE OR the measured best-safe reduction with a cited cold-cache reason; the AFTER run MUST show ZERO regression on the pre-build sweep + meta-test mutation sweep + propagation gates + a strong-model reasoning probe + a cache-warm proof (cache_read_input_tokens > 0) – cost reduction with quality regression is a \$. . . FAIL. The headline number is the *measured* reduction, never the design estimate (\$. . . / \$. . .). No measure may break/degrade any existing mechanism, and the rule is structured so none can. Composes \$. . . /. /. /. /. /. /. /. /. /. /. /. /. /. \$. . . / \$. . . Classification: universal (\$. . .).

Propagation gate CM-COVENANT-114-141-PROPAGATION (literal 11.4.141) + recommended gate CM-TOKEN-EFFICIENCY + paired \$. mutation (inject a pre-breakpoint volatile token → cache collapses → measured reduction falls below bar → gate FAILs). **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --skip-token-efficiency, --no-cache-governance, --assert-reduction-without-measuring, --tier-down-reasoning, --inline-all-governance, --tiktoken-estimate-OK flag.

\$. . – **Universal code-review mandate** – every change reviewed, always, no exception (User mandate, – –). Verbatim operator mandate: “ALL changes we do MUST pass through the code review step!!! ALWAYS!!!” EVERY change made to ANY repository this Constitution governs – without exception – MUST pass through an INDEPENDENT code-review step BEFORE it is accepted, committed, or built. NO change class is exempt: source code, fixes, tests, gates, meta-test mutations, documentation, doc-tooling, build/CI scripts, configuration, governance files (Constitution / CLAUDE / AGENTS / QWEN), conductor main-stream edits, sub-agent output, refactorors, single-line edits – if a diff exists, it gets an independent review. This is the ABSOLUTE form of \$. . (code-review agent gate after a batch, before pre-build sweep + main build): \$. . strips every implicit scoping \$. . ’s “after all fixes/changes/implementations are done” phrasing could leave open – no “just a doc edit”, no “just a one-liner”, no “the author already self-reviewed (\$. .)”, no “trivial change” carve-out. **Independence is load-bearing** – the reviewer MUST be structurally separated from the author (a dedicated code-review agent, subagent-driven by default per \$. . / \$. . , or a distinct human), NEVER the author re-reading their own work; \$. . ’s multi-pass self-evaluation is the AUTHOR-side discipline and PRECEDES (never satisfies) \$. . . **The review is**

itself anti-bluff (§ . / § . .) — a rubber-stamp “looks good” is not a review; it MUST genuinely analyse correctness, safety (no host § . / data § / target-hardware § . . regression), will-it-really-work (no solve-A-create-B), end-user behaviour (§ . / § .), and test-genuineness (§ .), its conclusions captured evidence per § . . / § . . , and it **iterates to a clean GO** per § . . — any finding (BLOCKING / nit / warning) re-arms the loop, acceptance only on ZERO new findings + ZERO warnings with rock-solid physical evidence. Honest boundary (§ . . .): a passing review does NOT replace § . . four-layer runtime-signature verification nor the § . . full-suite retest — it is one of MULTIPLE STRONG LAYERS, and the FIRST one every change crosses. Composes § . . / § . . / § . . / § . . / § . . / § . . / § . . / § . . / § . . / § . . / § . . . Classification: universal (§ . .) — the consuming project supplies its reviewer-dispatch mechanism + change-acceptance seam (commit wrapper / merge queue / PR gate) per § . . . Propagation gate CM-COVENANT-114-142-PROPAGATION (literal 11.4.142) + recommended gate CM-EVERY-CHANGE-REVIEWED (every accepted change carries a fresh independent-review-completed marker for its diff, produced by a reviewer distinct from the author, before acceptance/commit/build) + paired § . mutation (strip the literal → propagation gate FAILs; accept a diff with no independent-review marker → CM-EVERY-CHANGE-REVIEWED FAILs).

Canonical authority: constitution submodule Constitution.md § . . . Non-compliance is a release blocker. No escape hatch — no --skip-review , --no-review , --trivial-change-exempt , --doc-edit-exempt , --self-review-suffices , --review-after-commit flag.

§ . . – Real-user-journey mandate for video-streaming-app full-automation tests (User mandate, . . .). Verbatim operator mandate: “All video streaming apps ... require to choose some title and to press proper UI button to start or resume playing! Proper UI interaction to play exact show with proper content and subtitles is MANDATORY! Without it we just eventually play on and display sample, and that’s it mostly! THIS MUST BE ADDED as MANDATORY RULE regarding testing of any video streaming app in general with full automation tests! Universal in root constitution + a consuming project’s extensions.” Any full-automation test that asserts a video player / streaming application plays content MUST drive the REAL end-user journey through the app’s OWN UI – launch → BROWSE the actual catalog → choose a SPECIFIC title → press the real Play/Resume button → confirm THAT chosen content is genuinely playing, with its correct subtitles, on the intended routing target. A test that bypasses the journey with a sample / demo / built-in-loop clip, a deep-link / `am start -a VIEW` / intent shortcut, a synthetic or pre-staged stream, or any path that does NOT exercise the app’s own browse-select-play UI is a § . . . PASS-bluff at the user-journey layer: it validates ROUTING (that *something* reaches the display) while leaving the user-visible behaviour the operator cares about – “the show I picked actually plays” – unproven (the operator’s “we just eventually play on and display sample, and that’s it mostly” gap). The mandate (ALL hold): () **Real journey, not a shortcut** – launch → catalog browse → specific-title selection → real Play/Resume press through the app’s own UI (§ UI-driven), NEVER a deep-link / intent / sample / loop-clip shortcut; () **Chosen-content confirmation** – the PASS proves the SPECIFIC selected title is playing (not merely that pixels move on the target) via the § . . . liveness battery on the § . . . real-content path + the § . . . subtitle content-correctness oracle for the chosen title’s captions; () **Non-**

§ . . – Tracked/recorded-device availability-following mandate (User mandate, – –). Direct operator mandate: every device the project is tracking / following / recording (test / debug / manual-testing device, across every reachable transport – USB / wireless ADB / SSH / serial / network introspection API) MUST be availability-FOLLOWED – its connection state continuously monitored, any drop handled, never silently abandoned and never presented as a continuous recording. The § . . always-on recorder KNOWS a tracked device is absent (its per-device loop guards on the reachable state) but, lacking a following discipline, merely spins idle – no data captured, no offline event logged, no resume, no escalation – so the recording corpus presents a continuous timeline with a silent hole, a § . PASS-bluff at the recording-integrity layer (it claims continuous capture while no data exists for the gap). On a tracked device leaving its reachable state the system MUST, automatically: (a) **DETECT** the drop and **log an honest offline event** into the recording corpus (§ . . – a silent gap presented as continuous capture is a fabricated-continuity bluff; § . . – a silent recording gap defeats always-on recording); (b) **WAIT** for the device to return using the project's ALREADY-DEFINED reconnection timings – never invented numbers (§ . .), the SAME grace / reconnect / poll budgets the project's recovery path already uses; (c) **RE-ATTACH** – resume recording / tracking the moment the device returns and log an honest online / resume event; (d) **ESCALATE** to the project's sanctioned device-recovery path (per § . . feature class `device_recovery`) if the device does not return within the defined timeout – through the sanctioned, authorization-gated recovery entry point ONLY, never bypassing its gate and never performing a destructive recovery (e.g. a power-cycle) autonomously without that authorization (§ . . / § . . – high-blast-radius recovery is gated; while blocked the system keeps following the device, logging the

blocked-escalation honestly). A tracked-device drop that produces a silent corpus hole, a never-resumed recording, or a never-escalated permanent absence is the bluff this anchor forbids. Composes § . . . (always-on recording – § . . . closes its drop-handling gap) / § . . . (device_recovery sink-side positive evidence) / § . . . (honest offline / online events, reused-not-invented timings) / § . . . (watchdog children reaped on stop) / § . . . + § . . . (gated, non-autonomous escalation). Classification: universal (§ . . .) – the consuming project supplies its concrete tracking transport, device set, already-defined reconnection timings, and sanctioned recovery entry point per § Propagation gate CM-COVENANT-114-144-PROPAGATION (literal 11.4.144) + recommended gate CM-DEVICE-AVAILABILITY-FOLLOWED + paired § . . . meta-test mutation (strip the watchdog wiring / let an absence go unlogged → gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md § Non-compliance is a release blocker. No escape hatch – no --skip-availability-following , --silent-recording-gap-OK , --no-reconnect-wait , --invent-reconnect-timing , --skip-recovery-escalation , --autonomous-power-cycle-OK flag.

11 4 70 11 4 28

§ . . . – **Independent multi-angle impact-research per change** (User mandate, – –). For EVERY fix / change / new feature, INDEPENDENT impact-research agents (subagent-driven § . . . / § . . . , structurally separate from the author – § . . . self-eval PRECEDES, never satisfies – adversarial “refute-the-change” stance to defeat the documented LLM confirmation-bias failure mode) MUST research the change AND its connected/dependent features across a CLOSED SET OF EIGHT ANGLES – () correctness/logic, () regression (what existing working feature could break – via call-graph impact enumerating every direct + transitive caller and contract-dependent feature,

each mapped to its test), () latent/dangerous-code – the
 recents class (runtime-only failure, interface/ABI/contract
 mismatch, concurrency/data-race, resource lifecycle), ()
 security (taint/injection/secret/unsafe-API), ()
 performance (latency/memory/thermal under load vs baseline),
 () host/data/target-hardware safety per \$ /\$ /
 \$. . , () cross-feature interaction (shared
 state/timing/hardware contention), () business-logic
 conformance (matches the spec AND the connected features'
 contracts, not merely compiles). Forensic anchor (FACT,
 project-internal): the recents / -button-nav path shipped
 a latent AIDL-interface-contract mismatch that PASSED code
 review yet failed at RUNTIME ONLY (ANR on a recents-reaping
 gesture) because no review angle interrogated the interface
 contract / concurrency-lifecycle / silently-broken connected
 feature – the \$. . SOURCE→ARTIFACT→RUNTIME→USER-
 VISIBLE gap caught only after the fact; \$. .
 shifts that discovery LEFT. Each angle MUST name the TOOL(S)
 used + obtain the required DEPTH (the diff, the full changed
 unit, its declared contract, the spec section, every
 connected feature – NEVER the diff lines alone) + emit a
 captured-evidence conclusion per \$. . /\$. .
 (“looks fine” forbidden, \$. .); a genuinely-N/A angle
 is recorded NOT_APPLICABLE: <reason> per \$. . , never
 silently skipped. Output = a per-change impact-research
 REPORT (one section per angle: tool + evidence path +
 verdict) + a single GO/NO-GO that BLOCKS acceptance/commit/
 build on ANY unmitigated risk in ANY angle; the change is
 fixed/mitigated + affected angles re-researched, iterating to
 a CLEAN GO (zero unmitigated risk + zero warning) per
 \$. . . before the \$. . review gate.
 Honest boundary (\$. .): a GO proves cross-angle
 internal-consistency + bounded blast-radius – it does NOT
 replace \$. . runtime-signature verification nor
 \$. . full-suite retest; it is one of MULTIPLE
 STRONG LAYERS, the FIRST research pass every change crosses.

ABSENT – RED-on-broken then GREEN-on-fixed, both captured, on
 a clean target per § . . /§ . ,
 validated-first-after-redeploy per § . ,
 deterministic per § . ; no separate happy-path test
 substitutes (the § . /§ . PASS-bluff).
(STEP.. – EXTEND-TO-ALL-CASES, mandatory per-fix) NEW
EMPHASIS (**D**) immediately after STEP the test MUST be
 FANNED OUT across the full case-space of the SAME
 functionality – all flows, valid + invalid + boundary
 (§ . . empty/max/off-by-one) + concurrent
 (§ . contention) + failure-injection
 (§ . chaos) + topology variants (§ .) –
 confirming no issue of any kind, with PROVABLE enumerated
 coverage per § . (a listed case-set with per-
 case outcome, never “no other issues found”); a REQUIRED
 step, NOT deferred to a release-cycle discovery sweep and NOT
 reduced to a single guard; each user-visible case carries
 rock-solid physical evidence per § . + is
 registered into the § . standing regression-
 guard suite; a newly-discovered fan-out defect triggers
 § . test-interrupt + re-enters STEP .. (ANTI-
 BLUFF, all steps) every PASS is rock-solid CAPTURED physical
 evidence per § . (\$.. /\$. /
 \$..) – metadata-only / config-only / absence-of-
 error / grep-without-runtime PASS forbidden (§ . /
 \$..); unclear validation method ⇒ deep-research-
 before-declaring-untestable per § . /\$. /
 § . ; an operator-found defect the green suite
 missed triggers § . Honest boundary
 (§ .): STEP .. reduces the unknown-known surface
 but does not prove zero remaining defects (§ .) –
 un-exercised cases stated as honest gaps, never silently
 implied clean. Composes
 § . /. /. /. /. /. /.
 § ./\$. Classification: universal (§ .).
 Propagation gate CM-COVENANT-114-146-PROPAGATION (literal)

11.4.146) + recommended gate CM-REPRODUCE-FIRST-THEN-EXTEND (every closed defect's fix carries a § 4.135 . . . reproduce-first polarity test with captured defect-characterisation [STEP 4.135] + its RED_MODE=0 GREEN confirmation [STEP 4.135] + an enumerated per-functionality extend case-set registered into the § 4.135 . . . suite [STEP 4.135]; a closure with the reproduce→confirm pair but NO enumerated extend case-set FAILs) + paired § 4.135 . . . meta-test mutation (strip the literal → propagation gate FAILs; close a defect with only the reproduce→confirm pair and no extend-case-set → CM-REPRODUCE-FIRST-THEN-EXTEND FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md § 4.135 Non-compliance is a release blocker. No escape hatch – no --skip-reproduce-first , --fix-without-red , --skip-extend-to-all-cases , --reproduce-confirm-suffices , --defer-extend-to-release-sweep , --single-guard-suffices , --no-defect-characterisation flag.

§ 4.135 . . . – Crashed-agent respawn-until-complete + no-work-loss registry mandate (User mandate, § 4.135 . . .). Verbatim operator intent: “any agent that crashed because of something MUST BE respawned and finish its work at some point! We MUST NOT lose any work, forget about or have it corrupted!” Forensic case study (FACT): dispatched subagents died on TRANSIENT causes (API Error: Server is temporarily limiting requests rate-limit killed at once, API Error: socket connection closed unexpectedly killed , one left PARTIAL edits – two source files written, the dependent SystemUI sliders + doc + test NOT done), each re-dispatched by pure conductor vigilance with NO mechanical guarantee – the moment conductor context is lost / compacted / the conductor itself crashes, an in-flight-but-dead work unit is silently dropped (lost / forgotten / corrupted). Every dispatched agent/subagent MUST be tracked through its full lifecycle so a crash NEVER loses, forgets, or corrupts its work; a crashed agent is NOT a completed agent (§ 4.135 . . . – “it probably finished before dying” is

a forbidden guess; abnormal termination is positive evidence the unit is OPEN). The mandate (ALL hold): (a) **durable agent REGISTRY** – every dispatched agent has an append-only machine-readable registry entry (stable id, task + \$. . . file-scope, declared output path(s) + \$. . . / \$. . . evidence dir, dispatch timestamp, status from the CLOSED SET {dispatched | in-flight | crashed | respawned | complete}), reusing the \$. . . sync substrate (JSONL event stream + atomic status snapshot; “an entry with no dispatch-event cannot show complete”); the registry is the SINGLE SOURCE OF TRUTH for “is this work owed?” and an unregistered dispatch is itself a violation; (b) **mechanical CRASH-DETECTION + RESPAWN-until-complete** – any abnormal termination (transient rate-limit/socket-close, any non-completion exit, or a terminal error after the runtime’s own retries are exhausted) flips the entry to crashed + keeps the unit OPEN, and the unit is respawned (fresh agent claims the same id + scope + preserved state) and re-respawned until an agent reaches complete ; respawn is the safe/reversible/ bounded decision taken autonomously per \$. . . (never blocks the loop for a human), transient causes use the project’s ALREADY-DEFINED backoff budgets (never invented, \$. . .), a deterministically-reproducible non-transient terminal error is investigated per \$. . . before further respawn (never a blind retry loop); (c) **PARTIAL-STATE preserve → \$. . . -check → resume-or-clean-restart** – the crashed agent’s uncommitted edits + output doc are PRESERVED (never silently discarded → lost, never blindly committed → corruption), the respawn runs the \$. . . quiescence check on the preserved tree (every modified file accounted-for vs scope, no mutation/ // always pass / _mutated_* residue, no half-written/torn artifact), then EITHER RESUMES idempotently when it passes OR CLEAN-RESTARTS from a known-good base (\$. . . pre-op backup, reversible \$. . .) when inconsistent – nothing lost, nothing corrupted; per-agent git worktree isolation (\$. . . L /

§ . . – Workable-item integrity (status+type+id) + comprehensive structured description + bidirectional external-tracker sync + **BLOCKED** unblock-choices mandate (User mandate, . . . – . . .). BINDS + STRENGTHENS the workable-item discipline (§ . . . status / § . . . type / § . . . id / § . . . Operator-blocked / § . . . description-clarity / § . . . + § . . . SQLite SSoT / § . . . docs_chain) into ONE integrity contract spanning DB docs external tracker, adding the operator's three emphases: (D) **no item without a valid status + valid type + stable id – on ALL three surfaces** (an item missing any of the three in the DB, the rendered docs, OR the external tracker FAILS the validator – release-blocker; the id is the cross-surface binding key); (D) **comprehensive structured description per item – WHAT** it is (§ . . . ≥ -word/≥ -char clear meaning) + **HOW** it manifests + **HOW** to reproduce (§ . . . /.) + ACCEPTANCE CRITERIA (§ . . . /. captured-evidence verdict), in the § . . . DB description column AND the docs AND the tracker, never a stub/§ . . . -anti-pattern fragment; (D) **BLOCKED items carry WHY + enumerated unblock CHOICES** – tightens § . . . : every Operator-blocked item (incl. Blocked / BLOCKED documented alias normalised to canonical Operator-blocked , never a silent fork § . . .) MUST enumerate the closed list of decisions/actions that would unblock it ([A]...[B]...[C]... , mirroring § . . . 's – -option shape); a BLOCKED item with no enumerated choices FAILS; (D) **regular never-missed bidirectional DB docs .. tracker sync** – the § . . . /. git-tracked SQLite DB is the SSoT, docs + tracker DERIVED; full sync.. (db-to-md / md-to-db byte-identical round-trip § . . . + tracker push) runs regularly + on every change, § . . . drift-proof fingerprint (sha of the sorted item keyset, NOT mtime) gates freshness, § . . . docs_chain-bound so drift is mechanically

caught not vigilance-dependent; (**D**) **generic external-tracker push** carries statuses (collapsed onto the tracker's native set per § 11.4.148 / 149). When fixed, precise value preserved in a header, never lost), types, assignee (§ 11.4.148 participant handle, UNSET defaults from a project env var, never hardcoded/logged § 11.4.148), and sub-tasks, IDEMPOTENT (dry-run-then-real, match-by-stable-key [**<ID>**] prefix / id custom-field, present⇒UPDATE/absent⇒CREATE, rate-limited, credential-redacted, sink-side created=N updated=M failed=0 .. proof § 11.4.148), the MACHINERY project-agnostic § 11.4.148 (consumer registers its tracker/list/field-map at runtime). Anti-bluff § 11.4.148 : every sync + validator pass carries captured evidence § 11.4.148 / 149 ; honest boundary § 11.4.148 – guarantees well-formed items + agreeing surfaces, NOT that the underlying work is correct (still crosses § 11.4.148 / 149). Composes § 11.4.148 / 149 . Classification: universal (§ 11.4.148) – consumer supplies its DB path, id prefix, tracker service + list/board id + field map + default-assignee env var, docs_chain context per § 11.4.148 . Propagation gate CM-COVENANT-114-148-PROPAGATION (literal 11.4.148) + recommended gates CM-ITEM-INTEGRITY-STATUS-TYPE-ID / CM-ITEM-COMPREHENSIVE-DESCRIPTION / CM-BLOCKED-UNBLOCK-CHOICES / CM-TRACKER-SYNC-IDEMPOTENT + paired § 11.4.148 mutation (gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md § 11.4.148 . Non-compliance is a release blocker. No escape hatch – no --item-without-status , --item-without-type , --item-without-id , --stub-description-OK , --blocked-without-choices , --skip-tracker-sync , --one-way-sync-OK , --non-idempotent-tracker-push , --hardcode-assignee flag.

§ 11.4.148 – Per-workable-item testing-diary mandate (User mandate, – –). Every workable item MUST carry an append-only TESTING DIARY – a chronological record of every test run against it – distinct from § 11.4.148 .

item_history (lifecycle STATE transitions) + \$. .

Reopens (reopen cycles): the diary records TEST EXECUTIONS (which may or may not change status). The mandate (ALL hold):

(a) a test_diary table additive to the \$. . /.

SQLite SSoT, one row per run soft-keyed to the item_id, capturing ISO- -UTC date_time + tested_by (closed set User|Operator|AI-agent|HelixQA) + result (\$. . PASS|FAIL|SKIP + detail). + in-depth observations (long markdown, facts or explicit UNCONFIRMED: / PENDING_FORENSICS: \$. . , the background a future fix needs) +

action_taken + status_changed + from/to (did this run change status + WHY) + \$. . evidence_path + feature_class , with a SCHEMA CONSTRAINT making a PASS row WITHOUT a non-empty evidence path impossible (a PASS-bluff rejected by the schema itself); (b) BOTH an in-depth diary doc + a derived at-a-glance summary VIEW (total/pass/fail/skip + last-verdict + last-run + status-changes + distinct testers/feature-classes, DERIVED never duplicated \$. .) feeding \$. . risk-ordering; (c) four-format per-item Diary / Diary_Summary exports \$. . + \$. . - drift-proof-fingerprinted (sha of the sorted diary keyset) + \$. . docs_chain-bound so a diary row not exported/pushed FAILs a gate (never-missed); (d) external-tracker SUB-TASK model – the item task’s description stays the item (\$. . D , NOT polluted with diary text), each run a CHILD sub-task (type task) with a {TODO|In-progress|Completed} lifecycle (collapsed per \$. . /.

), observations + action + an Evidence: line (path not raw artefact \$. . /.

) + a diary-entry idempotency key, reusing the \$. . D idempotent rate-limited credential-redacted sink-side-proven push, mapper, project-agnostic \$. . ; (e) MINIMAL-LLM deterministic bash/Go tooling (zero LLM in the data path – the observations prose is authored by whoever ran the test; tooling only stores/renders/pushes/validates), % test-covered \$. . (unit + integration-against-the-real-

tracker with an honest \$. . SKIP-when-token-absent
never a faked PASS + export + HelixQA Challenge + paired
\$. . . mutation) so a diary PASS-bluff is mechanically
impossible. Honest boundary \$. . . – the diary
guarantees a complete auditable per-item test-history, NOT
that any single run’s verdict is correct (rests on its own
\$. . . /. /. evidence). Composes
\$. . . /. /. /. /. /. /. /. /. /.
\$. . Classification: universal (\$. .) – consumer
supplies its DB path, diary doc layout, export formats,
tracker sub-task field map, docs_chain context per
\$. . . Propagation gate CM-COVENANT-114-149-PROPAGATION
(literal 11.4.149) + recommended gates CM-TEST-DIARY-SYNC /
CM-DIARY-PASS-REQUIRES-EVIDENCE + paired \$. . mutation
(gate-code = separate work item). **Canonical authority:**
constitution submodule Constitution.md \$. . . Non-
compliance is a release blocker. No escape hatch – no --skip-
testing-diary , --diary-without-evidence , --no-diary-summary , --
diary-without-tracker-subtask , --llm-in-diary-data-path , --diary-
export-optional flag.

\$. . . – **Mandatory deep multi-angle web research
per change/issue, before declaring fixed or structural (User
mandate, – –)**. Verbatim operator mandate:
“For every single issue we fix or improvement we make –
besides tight systematic-debugging, fixing, code review by
independent agents, comprehensive tests – we MUST ALWAYS do
deep web research from various angles! No matter how big/
small/simple/complex, dig deep the internet for articles,
technical documentation, APIs, open-source code – EVERYTHING
that can help make the best possible solution OR confirm we
don’t have a more serious problem we’re unaware of! We MUST
do everything possible to STOP the constant issue-reopening
and finally start closing items as fixed+working! Do this
ALWAYS in parallel with the main work stream!” For EVERY
fix / improvement / change / closure – no matter how big,
small, simple, or complex – the agent MUST, IN ADDITION to

tight systematic-debugging (§ . . .), the fix,
 independent-agent code review (§ . . . /
 § . . .), and comprehensive multi-layer tests
 (§ . . . (b) / § . . .), perform a DOCUMENTED **deep
 multi-angle web research pass** digging the internet from
 VARIOUS angles – articles, official + vendor technical
 documentation, API references, standards, issue trackers,
 maintainer guidance, reusable open-source code – to BOTH (i)
 discover the best possible solution AND (ii) confirm there is
 NOT a more serious underlying problem the team is unaware of.
 Forensic case study (FACT, – –): a subtitle-
 on-secondary-display goal verdicted **Won't-fix: structurally-
 impossible** (§ . . . secure-surface pixel-blanking)
 was OVERTURNED by a deep multi-angle research pass that
 surfaced the real OCR-of-the-PRIMARY-display path – a “we
 can’t” became a shipped capability; the canonical anti-
 pattern of a structural verdict reached for want of research.
 The mandate (ALL hold): (A) **No closure-as-fixed/structural
 WITHOUT a documented deep-research pass** – no item may be
 marked **Fixed / Implemented / Completed** (§ . . .) OR
 classified **structurally-impossible won't-fix** (§ . . .)
 until a deep multi-angle pass is documented; § . . .
 makes § . . . 's pass UNCONDITIONAL (every item, however
 trivial, at the closure AND structural-verdict gates). (B)
Multiple angles, not a single lookup – ≥ 11 genuinely-
 distinct angles (official-docs / standards / known-bug-
 trackers / alternative-approach / failure-mode / security /
 performance / platform-constraint / open-source-precedent) so
 a single-source confirmation-bias miss (§ . . .)
 cannot pass; a one-link drive-by is NOT deep multi-angle. (C)
Confirm-no-bigger-problem, not just find-a-fix – explicitly
 seek evidence the fix does not mask a deeper defect AND the
 closure/verdict is genuinely safe; “we found nothing worse”
 requires the enumerated search (§ . . .). (D) **Latest-
 source + cited** – LATEST authoritative versions per
 § . . . (never training-data/memory), each cited by

URL + access date in the research artefact AND the closure
commit footer (Deep-research <date>: <urls> OR the literal NO
external solution found – original work per \$. .). (E)

Reopen-breaking is the PURPOSE – STOP the constant

Fixed Reopened churn ($\$ \dots / \$ \dots$); a reopen whose root cause a deep pass would have surfaced is a $\$ \dots$ miss; composes $\$ \dots$ (demotion-evidence) + $\$ \dots$ (structural verdict now REQUIRES

the cited-authorities pass). (F) ALWAYS in parallel with the main stream— background subagent-driven (\$. . . / \$. . . / \$. . . / \$. . .)

concurrent with main fix/build/test work, NEVER serialising it or stalling the loop (\$ 11.1.1 / \$ 11.1.2 / \$ 11.1.3 / \$ 11.1.4). (G) **Apply ASAP to every workable item** – retroactively + going forward across the \$ 11.1.1 / \$ 11.1.2 SSoT; items closed without a documented pass are re-audited in the \$ 11.1.1 / \$ 11.1.2 release-gate sweep. Honest boundary

(`$ . . .`): the pass reduces the unknown-unknown surface + breaks the most common reopen causes – it does NOT prove zero remaining defects (`$ . . .`) and does NOT replace

```
$ . . runtime-signature verification,
$ . . / $ . . . . review, or $ . . .
retest; it is the research layer every fix/closure/
```

structural-verdict additionally crosses, and “we probably don’t have a bigger problem” without the enumerated multi-angle search is a guess (§ 1.1.1), never a finding.

[illegible]

universal (§ 1.1) – the consuming project supplies its concrete research corpora, angle set, item tracker, and

closure-commit-footer convention per §

Propagation gate CM-COVENANT-114-150-PROPAGATION (literal 11.4.150) + recommended gate CM-DEEP-RESEARCH-PER-ISSUE (every closed/structural-verdicted item carries a documented multi-angle deep-research artefact + cited-source closure footer, run in parallel, before the closure/structural verdict is accepted) + paired § . . . mutation (strip the literal → propagation gate FAILs; close an item or reach a structural verdict with no documented multi-angle research artefact / cited footer → CM-DEEP-RESEARCH-PER-ISSUE FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md § Non-compliance is a release blocker. No escape hatch – no --skip-deep-research , --single-source-suffices , --trivial-change-no-research , --close-without-research , --structural-without-research , --serialise-research , --research-from-memory-OK flag.

§ . . . – **Project-prefixed release-tag/version-naming mandate** (User mandate, – –). Verbatim operator mandate: “Every release tag and version name we create – on the main repository and on every Submodule we own – MUST be prefixed with the project’s release prefix, e.g. myproject-1.0.0-dev-0.0.1 . The prefix MUST come from HELIX_RELEASE_PREFIX in our .env if it is set, otherwise from the lowercased project root directory name. The SAME prefix MUST be used across the main repo and all owned Submodules in one release so a release is greppable across every repository.” Every release tag AND every version name created on the main repository AND on every owned-by-us submodule (§ . . .) MUST be prefixed with the project’s release prefix, form <PREFIX>-<version> (canonical example: myproject-1.0.0-dev-0.0.1), so a release is identifiable + greppable across every repository it spans (one git tag -l '<PREFIX>-*' enumerates the whole release surface). **Prefix resolution order (closed-set, deterministic – § . . . no-guessing):** () HELIX_RELEASE_PREFIX from the project’s .env – authoritative when set; .env is git-ignored per

§ 11.4.14 and the variable is documented in the tracked `.env.example` (a § 11.4.14 re-obtain mechanism, never committed); () fallback = the lowercased snake_case form of the project root directory name (no spaces) per § 11.4.14. – used whenever the env var is unset/empty, so a prefix is ALWAYS resolvable from the checkout with zero operator input. **The prefix MUST be IDENTICAL across the main repo and all owned submodules within a single release** – a release that tags the main repo `<PREFIX>-1.2.0` while tagging an owned submodule with a different/unprefixed value is a § 11.4.14 violation (the cross-repo grep no longer enumerates the release). Version codes increment monotonically within the prefix (`<PREFIX>-...-0.0.1` → `<PREFIX>-...-0.0.2` → ...), never reset, never skipped. Honest boundary (§ 11.4.14): the prefix guarantees a release is identifiable + uniform across every repository, NOT that its contents are correct – the tag is still created only after the § 11.4.12 full-suite retest GREEN and reaches every upstream via the § 11.4.14 merge-onto-latest-main path (NEVER a force-push), fanned out per § 11.4.14. Composes `$ . / $ . . / $ . . / $ . . / $ . . / $ . .` (the release-scope terminal condition is a published, prefixed tag). Classification: universal (§ 11.4.14) – the consuming project supplies its concrete prefix value + the `HELIX_RELEASE_PREFIX` env var per § 11.4.14. Propagation gate `CM-COVENANT-114-151-PROPAGATION` (literal 11.4.151) + recommended gate `CM-RELEASE-PREFIX-NAMING` (every release tag/version on the main repo + every owned submodule carries the resolved `<PREFIX>-` prefix, identical across the release) + paired § 11.4.14 meta-test mutation (strip the literal → propagation gate FAILs; create an unprefixed or differing-prefix release tag → `CM-RELEASE-PREFIX-NAMING` FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` § 11.4.14. Non-compliance is

a release blocker. No escape hatch – no `--no-release-prefix`,
`--unprefixed-tag`, `--prefix-optional`,
`--differing-submodule-prefix` flag.

§ . . – **Crashlytics-recorded-data continuous monitoring + systematic-debug + regression-test-coverage mandate** (User mandate, – –). Verbatim operator intent: “For every project that has Firebase Crashlytics enabled / wired, we MUST continuously monitor ALL of the Crashlytics-recorded data – crashes, ANRs, performance traces, and non-fatals – systematically debug each, fix and improve, and cover everything with validation and verification tests. This MUST be checked regularly, with no false results and no bluff of any kind!” Every project that has Firebase Crashlytics enabled/wired (SDK linked into a shipping artifact, crash+non-fatal+ANR reporting active) MUST treat the Crashlytics console as a first-class captured-evidence channel from real end-user devices and continuously process every datum it records – an open Crashlytics issue with a green test suite is a § . . PASS-bluff at the field-telemetry layer (a real user hitting a broken feature while the suite reports green). STRENGTHENS+COMPLETES § . . : § . . owns the periodic REVIEW + dedup + Issue-creation pass that SURFACES Crashlytics/Analytics/Performance findings; § . . owns what happens to each surfaced item AFTER – the systematic-debug → fix/improve → regression-test-coverage lifecycle that drives it to a proven regression-immune closure (§ . . finds it; § . . fixes it + proves it stays fixed; both mandatory, neither substitutes). The mandate (ALL hold): () **continuous monitoring of ALL four surfaces** – fatal crashes, ANRs, performance traces/regressions, AND non-fatals (skipping any one is a PASS-bluff; non-fatals are the silent class – every catch/fallback/recovered-error path is a real degraded UX and MUST be triaged, not ignored because the app did not crash); () **systematic-debugging of each (reproduce-before-fix)** per § . . Iron Law + § . .

RED-baseline-on-the-broken-artifact (the recorded stacktrace/ANR-trace/non-fatal-context IS the \$. . /. captured defect-present evidence) BEFORE the fix – a fix without a prior reproducing test is a \$. . /. violation; unreproducible ⇒ deep-research-before-declaring-untestable \$. . /. ; () a **fix/improvement for every confirmed issue** against its proven root cause (\$. . /. /.), “acknowledged in console” is not a fix, muting a recurring issue without a tracked rationale is a \$. . /. violation; () **validation+verification regression-test coverage per closed issue** – in the SAME commit as the fix register a permanent \$. . regression guard (\$. . polarity test: RED_MODE=1 captures the recorded defect on the pre-fix artifact, RED_MODE=0 standing GREEN guard asserting ABSENT) exercising the same user-reachable path the stacktrace identifies, with rock-solid captured evidence \$. . /. /. /. + a closure log citing the console issue id/URL + root-cause + fix commit + the validation/verification test paths; a Crashlytics issue marked resolved WITHOUT a falsifiable regression test is FORBIDDEN (the silent-recurrence vector, \$. . operator-escape class applied to field telemetry); () **regular cadence** reusing the \$. . five-trigger set (pre-build/pre-flash/pre-distribute/pre-tag blocking; daily + post-deployment-burn-in non-blocking) – a one-time sweep never repeated is a violation; () **no false results, no bluff** (\$. . /. /. – “no new issues” requires the enumerated monitored-surfaces+window \$. . ; “fixed” requires the RED→GREEN flip \$. . ./. ; a guard whose \$. mutation does not FAIL it is a bluff gate). Honest boundary \$. . : processing every recorded issue breaks the silent-recurrence vector – it does NOT prove zero remaining field defects nor replace \$. . /. ; an issue is “closed” only when its guard is GREEN on a clean artifact, never when the console mark is flipped.

Classification: universal (§ . .) – the consuming project supplies its Crashlytics handle, console-access credential (§ . . , never logged), severity table, and per-issue closure-log path per § . . ; the reference project-level instantiation is a consuming project's § .0 (Crashlytics-Resolved Issue Coverage – per-issue validation test + Challenge .test + `..lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md` closure log) + § .AC (Comprehensive Non-Fatal Telemetry – every catch/fallback records a non-fatal with triage context). Composes

§ . /. /. /. /. /. /. /. /. /. /. /. /.

§ . . . Propagation gate `CM-COVENANT-114-152-PROPAGATION` (literal `11.4.152`) + recommended gate `CM-CRASHLYTICS-ISSUE-FULLY-COVERED` (every closed Crashlytics issue carries a systematic-debug root-cause record + a registered

§ . . regression guard + a closure-log entry citing the console issue id/URL + the validation/verification test paths; a console-resolved issue with no falsifiable regression guard FAILs) + paired § . meta-test mutation (gate-code = separate work item). **Canonical authority:**

constitution submodule `Constitution.md` § . . . Non-compliance is a release blocker. No escape hatch – no `--skip-crashlytics-monitoring` , `--monitor-crashes-only` , `--skip-non-fatals` , `--resolve-without-regression-test` , `--console-mark-is-fixed` , `--monitor-once` , `--mute-without-rationale` flag.

§ . . – **Comprehensive per-feature Status + Status_Summary** document set with mandatory video-recording confirmation (User mandate, – –). Every project MUST maintain, under `docs/features/` , a comprehensive feature Status document set (`Status.md` + its § . . `Status_Summary.md` companion) enumerating EVERY system component, EVERY client app/binary/surface (TUI / CLI / Web / desktop / mobile / API / gRPC / library / submodule / infrastructure), and EVERY feature – including features ported from any incorporated CLI-agent / submodule catalogue

(\$) – with NO single feature left out. () **Total categorized coverage** – per-feature table organized Component→Category, reconciled against the actual codebase (a code-present feature missing from the table, or a row with no code, is a \$ /\$ bluff). () **Per-feature fields** – Component / Feature / Category / Implementation / Wiring (genuinely end-user-reachable, not merely compiled, \$) / Real-use / Tests-coverage (four-layer \$ (b)) / Validation (PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED per \$) / **Video-recording confirmation** (path to the real-use video or honest gap marker). (. . .) **Mandatory per-feature real-use video** – every user-visible “confirmed” claim backed by a recorded real-use video of a genuine end-user scenario (real prompts → real LLM/service responses → real results), NEVER a frozen/stale frame (\$), NEVER faked/mocked/demo-loop/bluff response or LLM error passed off as success (\$ /\$), stored at the project-declared recording path (\$); a confirmed row with no real video or a bluff video = \$ PASS-bluff; autonomous-infeasible ⇒ honest \$ /\$ SKIP + tracked migration item, NEVER a faked confirmation. () **Video-analysis remediation loop** – every video analysed; any defect it surfaces triggers \$ systematic-debug → fix → \$ retest → re-record → clean GO per \$ before “confirmed” (a video exposing a broken feature is a \$ test-interrupt). () **Always-in-sync** – \$ -class roster/corpus-backed, \$ docs_chain-bound + \$ drift-proof fingerprint (sha of sorted feature-key roster AND sorted video-artefact roster, NOT mtime), re-syncs out-of-the-box; stale = violation. () **Four-format export** – HTML + PDF + **DOCX** (this doc class ADDS DOCX to the \$ HTML+PDF set; other classes unchanged), in sync per \$ () Follows \$ /. /. /. /. /. () **MP**

format REQUIRED. All video confirmations MUST be in `.mp4` format (H.). Window-specific capture ONLY (\$. . (A)). Vision validation REQUIRED (\$. . (D)). `.cast` files are supplementary only. Honest boundary \$. . – guarantees a complete video-confirmed always-synced ledger, NOT per-feature correctness (rests on each row's \$. . /. /. evidence) and NOT a \$. . retest substitute. Classification: universal (\$. .). Composes \$. . /. /. /. /. /. /. /. /. /. /. /. \$. . Propagation gate `CM-COVENANT-114-153-PROPAGATION` (literal `11.4.153`) + recommended gates `CM-FEATURE-STATUS-COMplete` + `CM-FEATURE-STATUS-VIDEO-CONFIRMED` + paired \$. mutation.

Canonical authority: constitution submodule `Constitution.md` \$. . . Non-compliance is a release blocker. No escape hatch – no `--skip-feature-status`, `--feature-without-video`, `--frozen-video-OK`, `--bluff-response-video-OK`, `--skip-video-analysis`, `--feature-ledger-incomplete-OK`, `--no-docx-export`, `--allow-feature-ledger-drift` flag.

\$. . – **Window-scoped capture + fresh-corpus rotation for feature/QA recordings** (User mandate, . . . –). Verbatim: “recording you perform does record the window containing apps and services, not the whole desktop or monitor screen!” + “All old recording files MUST BE removed when new one starts!” Refines \$. . /. /. /. recording discipline with two capture-hygiene invariants. (A) **Window-scoped, NOT whole-screen** – every feature/QA video MUST capture ONLY the window/surface of the app/service under test (GUI window / CLI-TUI terminal pane / web tab-viewport / device-emulator-simulator frame), NEVER the whole desktop/monitor or unrelated windows; whole-desktop capture leaks operator-private content (\$. . /.), dilutes the \$. . liveness/freeze oracle, and breaks the \$. . OCR/

ROI content oracle. Target the window/region by stable identity (window id/title, device serial, browser context, tmux target) per § . . . – never a fixed full-screen device index. Platform genuinely cannot capture below whole-screen ⇒ honest § . . . SKIP + tracked migration item, never a whole-screen pass-off. **(B) Fresh-corpus rotation** – when a new recording run for a scope begins, the agent’s OWN prior in-scope stale recordings at the raw recording path MUST be removed FIRST so the live corpus reflects the current run (§ . . . not-stale + § . . . roster-freshness). Honest boundary (§ . . . + § . . .): “remove old” = the agent’s own prior recordings for the SAME scope/project ONLY – NEVER another project’s/scope’s/operator-authored files; uncertain ⇒ surface, don’t delete (§ . . .); committed docs/qa/<run-id>/ evidence (§ . . .) is the durable record, NOT rotated. Classification: universal (§ . . .). Composes § . . . /. /. /. /. /. /. /. /. /. /. Propagation gate CM-COVENANT-114-154-PROPAGATION (literal 11.4.154) + recommended gate CM-WINDOW-SCOPED-FRESH-CORPUS-RECORDING + paired § . . . mutation.

Canonical authority: constitution submodule Constitution.md § Non-compliance is a release blocker. No escape hatch – no --whole-screen-capture-OK , --skip-window-scope , --keep-stale-recordings , --no-corpus-rotation , --full-desktop-recording flag. **(C) MP auto-conversion REQUIRED.** Any .cast file produced MUST be auto-converted to .mp4 via agg + ffmpeg immediately after capture. The .mp4 is the primary evidence; .cast is supplementary only.

§ . . . – **Project-name-prefixed feature/QA recording filenames** (User mandate, – –). Verbatim: “All recorded videos MUST START with prefix: the PROJECT NAME (ALWAYS USE THE PROJECT NAME). Project name MUST be obtained according to the constitution’s own project-name resolution.” Every recorded video the project produces – every feature/QA

real-use recording (\$. . .), every window-scoped capture (\$. . .), every always-on device recording (\$. . .), every raw or curated artefact at the project-declared recording path (\$. . .) + the committed docs/qa/<run-id>/ trail (\$. . .) – MUST have a filename that STARTS WITH the PROJECT-NAME prefix, ALWAYS; an unprefixed recording is a \$. . . violation (a multi-project/scope corpus on one host per \$. . . / . . . becomes un-greppable + un-attributable – the \$. . . identify-and-grep failure on the recording-corpus axis). **Prefix resolution (closed-set, deterministic – \$. . . , IDENTICAL to \$. . .):** () HELIX_RELEASE_PREFIX from .env (authoritative, git-ignored \$. . . , documented in tracked .env.example \$. . .) else () lowercased snake_case project-root dir name \$. . . – ALWAYS resolvable, zero operator input. SAME prefix for EVERY recording in a checkout so ls '<PREFIX>---'* enumerates the corpus; canonical form <PREFIX>---<feature-or-scope>---<run-id>.<ext> (--- delimits the prefix unambiguously); MUST equal the \$. . . - resolved release-tag prefix (divergence is itself a \$. . . violation – one project, one name). Honest boundary (\$. . .): the prefix guarantees attribution + greppability, NOT content validity (still \$. . . / . . . / . . .) and does NOT relax \$. . . 's window-scope/rotation (rotation removes the agent's OWN <PREFIX>---* only; foreign/operator files surfaced not deleted \$. . . / \$. . .). Classification: universal (\$. . .). Composes \$. . . / . . . / . . . / . . . / . . . / . . . / . . . \$. . . Propagation gate CM-COVENANT-114-155-PROPAGATION (literal 11.4.155) + recommended gate CM-RECORDING-PROJECT-NAME-PREFIX + paired \$. . . mutation.

Canonical authority: constitution submodule Constitution.md
\$. . . Non-compliance is a release blocker. No
escape hatch – no --no-recording-prefix , --recording-without-
project-name , --unprefixed-recording , --prefix-optional-for-
recording , --differing-recording-prefix flag.

\$. . . – All CI/CD automation (GitHub Actions /
GitLab pipelines / equivalents) MUST be disabled (User
mandate, – –). Verbatim operator mandate:
“Any GitHub actions or GitLab pipelines MUST BE disabled! Add
this critical mandatory rule / mandatory constraint into the
root constitution Submodule, commit and fetch all its changes
to all upstreams and make sure we respect and follow this
rule (we do apply it) ASAP!!!” Every repository this
Constitution governs – main repo, this constitution
submodule, every owned + nested submodule we author and push
– MUST ship with ALL server-side CI/CD automation DISABLED:
no push to any owned upstream may trigger a GitHub Actions
run, GitLab pipeline, or equivalent (Jenkins/CircleCI/Travis/
Drone/Woodpecker/Bitbucket/Azure, any on: push / schedule /
workflow_dispatch). GENERALISES + makes ABSOLUTE the
\$. . . . Layer- posture (remote CI DISABLED,
workflow preserved at a ...disabled-local-only non- .yaml name a
provider ignores) across ALL governed repos; enforcement
migrates to the LOCAL \$. . . git-hook ritual +
\$. . . pre-tag sweep, never a remote runner. ALL
hold: (A) zero active.. .github/workflows/*.yaml|yaml / .gitlab-
ci.yaml / .gitlab/** / equivalent at the ROOT of any governed
repo/submodule (the only place a provider executes); (B)
“disabled” = a push triggers ZERO runs – delete OR rename to
a non-trigger name (\$disabled / .disabled-local-
only); a live- on: + if:false workflow still queues runs, NOT
compliant; (C) ..scope = repos we author+push – vendored/third-
party nested configs below the root (AOSP external/** ,
prebuilts/** , vendored submodules) are INERT (a provider
never runs a non-root config), OUT of scope, MUST NOT be
mass-edited (\$. . . vendor-exempt); test = “does a

push to OUR upstream trigger a run?" yes⇒disable,
inert⇒document+leave (\$. . verify-not-assume); (D) no
new CI may be added (release blocker); (E) pre-push verify
git ls-files | grep -E '^\.github/workflows/.*\ya?ml\$|^\.gitlab-
ci\.yaml\$' empty for authored repos, \$. . -class
PreToolUse guard + gate enforce mechanically. Honest boundary
(\$. .): file-level disabling stops FILE-triggered
runs, NOT provider-side server settings (org-default required
workflows, branch-protection required checks, provider
scheduled exports) – the operator turns those off; the agent
documents what it cannot reach, never claims unachieved
completeness. Composes

\$. . /. /. /. /. /. /\$. .
Classification: universal (\$. .). Propagation gate
CM-COVENANT-114-156-PROPAGATION (literal 11.4.156) +
recommended gate CM-NO-ACTIVE-CI + paired \$. meta-test
mutation (strip the literal → propagation gate FAILs; add a
root .github/workflows/x.yml to an authored repo →
CM-NO-ACTIVE-CI FAILs). **Canonical authority:** constitution
submodule Constitution.md \$. . . Non-compliance is
a release blocker. No escape hatch – no --allow-ci , --enable-
workflow , --keep-pipeline , --remote-ci-OK , --ci-exempt flag.

\$. . – GEMINI.md maintained in lockstep with
CLAUDE.md / AGENTS.md / QWEN.md (User mandate,
– –). Verbatim operator mandate: “Make sure
with CLAUDE.md, AGENTS.md, QWEN.md we maintain GEMINI.md too!
Add this mandatory fact / rule to root constitution Submodule
we are inheriting / extending – CONSTITUTION.md, CLAUDE.md,
QWEN.md, AGENTS.md, GEMINI.md and other related relevant
files!” Forensic FACT (– –): when
\$. . /\$. . were authored,
Constitution/CLAUDE/AGENTS/QWEN carried the family through
\$. . but GEMINI.md had silently drifted to
\$. . – mandates (\$. . –)
never propagated – a \$. . propagation-bluff (a Gemini-
CLI agent reads a stale Constitution). GEMINI.md is a FIRST-

CLASS governance context carrier EQUAL to CLAUDE.md/
 AGENTS.md/QWEN.md, never optional/best-effort. ALL hold: (A)
 five-carrier lockstep – no governance change is complete
 until GEMINI.md carries it alongside the other three mirrors;
 GEMINI.md is added to the \$. . . propagation + cross-
 reference set explicitly; (B) no silent drift – GEMINI.md
 lagging the other mirrors' highest rule is a \$. . .
 violation (\$. . . -class), back-fill required; (C)
 equal status – GEMINI.md restates the SAME literal 11.4.N
 anchors the propagation gates require (\$. . .), fleet
 count INCLUDES GEMINI.md; (D) consumer projects' own CLAUDE/
 AGENTS/QWEN/GEMINI bind too (\$. . .). Honest boundary
 (\$. . .): the \$. . . – GEMINI.md back-
 fill is a tracked release-blocking remediation; claiming
 GEMINI.md “in sync” while the back-fill is incomplete is
 itself a \$. . . violation. Composes
 \$. . . /. /. /. /. /. /.
 \$. . . Classification: universal (\$. . .).
 Propagation gate CM-COVENANT-114-157-PROPAGATION (literal
 11.4.157 , GEMINI.md INCLUDED) + recommended gate CM-GEMINI-MD-
 LOCKSTEP + paired \$. . . meta-test mutation. **Canonical**
authority: constitution submodule Constitution.md
 \$ Non-compliance is a release blocker. No
 escape hatch – no --skip-gemini-md , --gemini-optional , --
 gemini-lag-OK , --four-carrier-suffices flag.

\$. . . – Intensive all-feature/flow/edge-case
 video-recording + read-the-screen content-verification
 mandate (User mandate, – –). Every project
 MUST be covered by intensive automated testing that exercises
 + RECORDS every feature/flow/use-case/edge-case (valid/
 invalid/boundary/concurrent/failure-injection \$. . .),
 each recording showing the feature GENUINELY WORKING with
 REAL results (real prompts→real responses→real outputs
 \$. . .) and NO false/simulated/stale/frozen result
 (\$. . . /. /.) – a recording showing an error/
 bluff/non-working feature is a FINDING (→\$. . . ()/

verification, challenge, and QA session that produces video evidence MUST comply with ALL of the following: (A) **Window-specific recording ONLY** – every video MUST record ONLY the target application window (Terminal pane, TUI app, browser tab, emulator frame), NEVER the whole desktop/monitor screen; use macOS `screencapture -l<window_id>`, Linux `xdotool + ffmpeg`, or equivalent; whole-desktop capture leaks operator-private content (\$ `xdotool getactivewindow`), dilutes the \$ `xdotool` liveness oracle, and breaks \$ `ocr` OCR/ROI; platform genuinely cannot capture below whole-screen => honest \$ `xdotool` SKIP + tracked migration item. (B) **MP format REQUIRED** – `.mp4` (H. `264`, `movflags +faststart`, `pix_fmt yuv420p`); `.cast` files are supplementary only; auto-convert via `agg + ffmpeg` per \$ `ffmpeg`. (C) **Project-name prefix REQUIRED** – filename MUST start with project name in snake_case from `HELIX_RELEASE_PREFIX` in `.env` (\$ `PROJECT_NAME`) or lowercased project root dir name (\$ `PROJECT_NAME`); format `<project_name>-<feature>-<timestamp>.mp4`; unprefixed = violation. (D) **Mandatory vision validation** – after EVERY recording, the agent MUST read the terminal output / video content and verify: (i) feature ACTUALLY WORKS, (ii) LLM responses are REAL, (iii) all tests show PASS, (iv) no “TODO implement” / “simulate” / “for now” patterns, (v) output demonstrates end-user working feature; MUST produce a verdict (PASS/FAIL) with evidence path (\$ `VIDEOPATH`). (E) **Terminal window cleanup** – after each recording, dismiss/close ONLY the Terminal window used for that recording (window-specific close via `osascript / xdotool`); MUST NOT close windows belonging to other processes. (F) **Real results ONLY** – every recording MUST show REAL working features; errors/empty output/simulated responses trigger fix→retest→re-record. (G) **Re-runnable evidence** – the command shown MUST be re-runnable to produce the same results. (H) **Fresh-corpus rotation** – remove agent’s own prior in-scope stale recordings FIRST (\$ `rm -rf recordings`). (I) **Content verification MANDATORY** – not duration-based – the

value of a recording is NOT its duration but its CONTENT; a recording MUST demonstrate the ACTUAL feature being used with REAL results; before accepting ANY recording, the agent MUST verify: expected output patterns ARE present (e.g., test PASS lines, API response data, feature-specific output), the feature ACTUALLY WORKS as demonstrated (not just “something ran”), LLM responses are REAL content (not simulated, not placeholder, not empty), every claim of “working” is backed by visible evidence; a 1-second recording showing a feature working correctly is MORE valuable than a 10-second recording of empty terminal; duration is NOT a proxy for quality. (J) **Expected-content specification REQUIRED** – before recording, the agent MUST specify what content SHOULD appear (expected patterns, expected test results, expected API responses); after recording, the agent MUST verify these patterns ARE present; if expected content is MISSING, the recording is REJECTED regardless of duration. (K) **Content-verification recording workflow** – mandated workflow: () SPECIFY expected content patterns, () RECORD the feature execution, () EXTRACT all text from the recording, () VERIFY expected patterns are present, () CHECK for simulated/placeholder content, () ACCEPT only if ALL patterns found AND zero bluffs detected, () REJECT and re-record if ANY pattern missing or bluff detected. (L) **Root cause analysis REQUIRED for rejected recordings** – when a recording is rejected (missing expected content, bluff detected, or empty capture), the agent MUST investigate WHY before re-recording per § 4.1.2.1; determine the root cause (timing issue, wrong command, tool failure, etc.) and fix it; simply re-recording without understanding WHY the first attempt failed is a § 4.1.2.1 violation. (M) **Real-time monitoring RECOMMENDED** – for complex features, use real-time monitoring that analyzes output DURING recording (not after); this catches issues immediately and allows corrective action before the recording completes. Classification: universal (§ 4.1.2.1). Composes

\$. . /. /. /. /. /. /. /. /. /. /.
 \$ Propagation gate CM-COVENANT-114-159-PROPAGATION
 (literal 11.4.159) + recommended gate
 CM-WINDOW-VIDEO-VALIDATED + paired \$. mutation. **Canonical
 authority:** constitution submodule Constitution.md
 \$ Non-compliance is a release blocker. No
 escape hatch – no --whole-screen-ok , --cast-only , --skip-
 vision-validation , --no-cleanup , --simulated-recording-ok , --
 unprefixed-recording flag.

\$. . . – **Vision-verified recording + HelixQA bridge
 mandate** (User mandate, – –). Compact summary:
 every video recording for feature/QA evidence MUST be
 processed through a vision/OCR pipeline that reads on-screen
 content and confirms expected results BEFORE acceptance; the
 recording system MUST provide a bridge feeding captured
 frames to HelixQA’s test infrastructure (or equivalent) for
 automated read-the-screen verification against SPECIFY-phase
 expected patterns (\$. . . (J)); capture frames at
 ≤ s intervals, run OCR/vision with self-validated analyzer
 (\$. . . ()), compare text against patterns,
 produce per-frame PASS/FAIL with evidence path, surface
 failures immediately for re-record (\$. . . (L)).
 Project-configured parameters per \$ Honest
 boundary: does NOT replace \$. . . quality analysis nor
 \$. . . runtime-signature; FLAG_SECURE uses
 \$. . . proxy oracle. Classification: universal
 (\$. . .). Propagation gate CM-COVENANT-114-160-
 PROPAGATION + recommended gate CM-VISION-VERIFIED-RECORDING-
 BRIDGE + paired \$. mutation.

\$. . . – **Rootless container runtime mandate** (User
 mandate, – –). Compact summary: every project
 MUST use Podman rootless (or equivalent) for ALL
 containerized workloads – rootful Docker/sudo FORBIDDEN
 unless \$. . . platform constraint documented;
 vasic-digital/containers Submodule (\$. . .) sole

orchestration layer via `pkg/boot / pkg/compose / pkg/health` ;
extend upstream per `$ . .` for missing capabilities;
integration tests boot infra on-demand. Honest boundary: does
NOT replace `$ . .` credentials nor `$ .`.
hygiene. Classification: universal (`$ . .`).
Propagation gate `CM-COVENANT-114-161-PROPAGATION` + recommended
gate `CM-ROOTLESS-CONTAINER-RUNTIME` + paired `$ .` mutation.

`$ . .` – **OpenDesign UI design system mandate (User mandate, – – `)`. Compact summary: every project with user-facing interfaces MUST use OpenDesign (<https://github.com/nexu-io/open-design>) as mandatory UI design system – NOT ad-hoc CSS; install as dependency, use token system for all color (light+dark from brand assets), typography, spacing, component tokens; extend upstream (`$ . .`) for unsupported patterns; every UI component ships light+dark + token-diff, regression tests via visual regression; no element overlap/font collision/label overlay. Honest boundary: does NOT replace `$ . .` functional testing nor `$ . .` / `.` UI testing. Classification: universal (`$ . .`). Propagation gate `CM-COVENANT-114-162-PROPAGATION` + recommended gate `CM-OPENDESIGN-UI-SYSTEM` + paired `$ .` mutation.**

`$ . .` – **Universal Media Validation (User mandate)**. Every recorded artifact MUST pass MEDIA VALIDATION pipeline – OCR/transcription/text parsing, compare vs SPECIFY-phase patterns (`$ . .` (J)), self-validated analyzer (`$ . .` ()), structured verdict, pinpoint on FAIL, REAL-TIME trigger option. Paired `$ .` mutation on golden-bad fixture. Classification: universal. Propagation gate `CM-COVENANT-114-163-PROPAGATION` + paired `$ .` mutation.

`$ . .` – **Universal Auto-Propagation Hook (User mandate)**. Every constitutional pull triggers `post_update_hook.sh` – detects changed files, registers skills/

MCP/hooks/scripts, logs summary. Consumer invokes after every pull. Classification: universal. Propagation gate CM-COVENANT-114-164-PROPAGATION + paired \$. mutation.

\$. . – **Universal Independent Verification Agent (User mandate)**. Every code/media/docs/config output passes independent verifier (\$. . /.), zero-finding GO (\$. .). CODE: review+build+mutations+runtime-signature. MEDIA: pipeline+genuine+format. DOCS: exports+revision. CONFIG: syntax+schema+leaks. Self-validated \$. mutation. Classification: universal. Propagation gate CM-COVENANT-114-165-PROPAGATION + paired \$. mutation.

\$. . – **REPEALED (operator decision,**
–). The Universal Semgrep static-analysis mandate is repealed – Semgrep is NO LONGER mandatory. Static analysis remains encouraged, not mandated. Scaffolding, submodules/semgrep , MCP/pre-commit/PATH wiring, docs_chain semgrep_status context, and the CM-COVENANT-114-166-PROPAGATION / CM-SEMGREP-WIRED gates removed. Anchor . . retired. See constitution Constitution.md \$. . .

\$. . – **Big-work-item feature work-stream lifecycle (User mandate,** –). Every BIG work item (new feature OR drastic/large fix) MUST be its own isolated feature work-stream – own sibling project copy, own feature/<slug> branch, own per-feature flashable builds + feature tags, SEPARATE from trunk until operator manually approves after testing, trunk merged INTO every stream regularly. (A) one big item = one sibling project copy (<project>_<slug>); small changes stay on trunk. (B) disk-feasible CoW/reflink clone MANDATORY (cp -a --reflink=auto on btrfs/XFS/ZFS/APFS, else git worktree); naive deep copies FORBIDDEN; per-stream out/ + shared ccache; ~ -disk verified (\$. .) before relied on (\$. .). (C) own branch + greppable tags (\$. . <base>-feat-<slug>), NEVER merged to trunk until \$. . GREEN on

clean target → \$. . . merge-first → \$. . .
 ff-only. (D) trunk merged INTO every live stream per trunk-
 tag/daily (merge, never rebase). (E) feature branch + tag
 cascades to EVERY touched submodule (\$. . .). (F)
 single-builder build QUEUE (\$. . . /. . . /. . .)
 + finite-device queue (\$. . .); idle out/ GC'd
 (\$. . .). (G) every change crosses full gauntlet
 (\$. . . /. . . /. . . /. . . + anti-bluff). (H)
 stream registry (\$. . . /. . .) + atomic
 .partial → rename + \$. . . resume + \$. . . retire-
 backup; bounded \$ /. . . Honest boundary
 (\$. . .): isolation+disk-feasibility+greppable-
 identity+merge-discipline, NOT correctness (still
 \$. . . /. . .). Classification: universal
 (\$. . .). Propagation gate CM-COVENANT-114-167-
 PROPAGATION (literal 11.4.167) + gates CM-FEATURE-WORKSTREAM-
 COW-CLONE / -NO-MERGE-UNTIL-APPROVED / -TRUNK-SYNC-CADENCE /
 -SUBMODULE-CASCADE + paired \$. . . mutations. **Canonical**
authority: constitution Constitution.md \$

11 4 65

\$. . . – Exported-document independent content +
 textual + full-visual validation mandate (User mandate,
 – –). Every generated/exported document (HTML/
 PDF/DOCX/any \$. . . export-scope format) MUST pass
 INDEPENDENT validation by a reviewer structurally separate
 from the generator (\$. . . /. . . , NEVER author self-
 check), on BOTH source AND every exported artifact, ALWAYS,
 across THREE layers: () CONTENT – export faithfully carries
 source intent+data, nothing dropped/truncated/garbled; ()
 TEXTUAL – human-readable, NO raw markup/diagram-source
 (Mermaid gantt / graph / flowchart / sequenceDiagram etc.)/
 unrendered code-fence leaking as body text (the exact
 – – raw-gantt-in-PDF failure); () FULL
 VISUAL – diagrams render as IMAGES not source, layout intact,
 no overlap/cut-off/garble, verified by RENDERING the export
 (pdftotext catches raw source + pdfimages confirms rendered
 images + pdftoppm → OCR confirms human-readable content) with


```

captured evidence..$. . . /. /. . Anti-bluff:
rubber-stamp ≠ validation; analyzer self-validated golden-
good/golden-bad $ . . ( ) (golden-bad seeded with
raw gantt MUST FAIL); iterate to clean GO $ . . .
Forensic FACT ( - 11 4 6 ): a “green” Mermaid fix
shipped raw gantt source as readable text into user PDFs
because validation grepped HTML for class="mermaid" and NEVER
checked the exported PDF – operator caught it
($ . . ). Honest boundary ($ . . ): confirms
the exported artifact is faithful/readable/visually-rendered
– does NOT prove source content is correct
($ . . /. ) nor replace the $ . .
mtime freshness gate (this is the READABILITY/FIDELITY layer
the mtime gate cannot see); FLAG_SECURE/un-rasterisable
surfaces document the gap $ . . + $ . .
proxy, never a faked PASS. Classification: universal
($ . . ). Composes
$ . . /. /. /. 11 4 168 /. /. /. /.
$ . . Propagation gate CM-COVENANT-114-168-PROPAGATION
(literal 11.4.168 ) + recommended gate CM-EXPORTED-DOC-VISUALLY-
VALIDATED + paired $ . . mutation. Canonical authority:
constitution Constitution.md $ . . .

```

fully-autonomous QA sessions § . .); () DDoS/load-flood (clean refuse or graceful degrade); () security (authz/injection/secret-leak § . . /crypto/CVE/default-deny + security submodule); () stress+chaos (load+contention+boundary+failure-injection, categorised recovery § . .); () concurrency/atomicity (no lost updates, transactional atomicity, idempotent replay); () race-condition/deadlock (race detector + lock-order analysis; no blocking op in shared-lock region §); () memory (leak census + peak-RSS ceiling + h/N-iter soak, no unbounded growth); () benchmarking/performance (p / p + throughput vs recorded baseline). The ONLY permitted absence is an honest § . . SKIP-with-reason, never a silent gap; four-layer § . . (b) enforcement + coverage ledger (§ . . /. .); missing required type or OPERATOR_ATTENDED_ONLY = release blocker. Strict enumerated expansion of § . . . Composes § . . /. /. /. /. /. /. /. /. /. /. /. /. /. \$. . Classification: universal (§ . .).

Propagation gate CM-COVENANT-114-169-PROPAGATION (literal 11.4.169) + recommended gate CM-MANDATORY-TEST-TYPES-COVERED + paired § . mutation. **Canonical authority:** constitution Constitution.md § . . .

§ . . – **Device-independent host-side rendered-UI visual-proof mandate** (User mandate, – –).

Compact summary: every change to ANY user-facing UI surface (screen/component/view/widget/layout/theme on any platform – Android-Compose, iOS-SwiftUI/UIKit, web, desktop, TUI) MUST be proven by DEVICE-INDEPENDENT host-side RENDERED PIXELS before it may be claimed correct – the real component rendered to a PNG ON THE HOST (NO device, NO emulator, NO running app) via a host-render harness (Compose→Roborazzi/Paparazzi on the JVM; web→Playwright/Storybook snapshot; SwiftUI→snapshot-testing; equivalent per stack), for EVERY relevant screen×state×{light,dark} theme, with the PNG validated by BOTH (i) golden image-diff (per-pixel/perceptual


```
constitution submodule Constitution.md $ . . . Non-  
compliance is a release blocker. No escape hatch – no  
--value-assertion-proves-ui , --skip-rendered-pixel-proof ,  
--token-equality-suffices , --device-offline-skip-visual ,  
--single-theme-only , --unvalidated-image-diff-OK ,  
--no-ocr-layout-oracle flag.
```

When in doubt

- Read constitution/Constitution.md for the canonical text of every rule.
- Cross-reference the sibling carriers (CLAUDE.md / AGENTS.md / QWEN.md) – they carry the same universal content for their respective agents.
- If still unclear, ask the operator. Do NOT guess. Do NOT bluff (\$. .).

\$. . – Mandatory comprehensive human-readable workable-item descriptions (User mandate, – –). Every workable item (ATM-NNN ticket) in the project’s workable-items database, Issues.md, Fixed.md, Issues_Summary.md, Fixed_Summary.md, and ALL derived documents MUST carry a comprehensive human-readable description that explains the item in plain, non-technical language suitable for non-developers (project managers, stakeholders, team leads, business partners). The description MUST be – sentences minimum, written so that ANY person – regardless of technical background – can understand: (a) WHAT the item is (the feature, fix, or task in simple terms); (b) WHY it matters (the user/business value); (c) HOW it works (the mechanism, without code references); (d) WHO benefits (the end user or stakeholder); (e) WHAT the expected outcome is (the measurable result). The description is MANDATORY – an item without a human-readable description is a \$. violation at the documentation layer. Descriptions MUST NOT use jargon, acronyms without expansion, code

references, or technical implementation details as the primary explanation. Technical details MAY appear as supplementary context AFTER the plain-language explanation. The `description` column in the SQLite database (`docs/workable_items.db`) is the SINGLE SOURCE OF TRUTH for item descriptions; all derived documents (Issues.md, Fixed.md, summaries, exports) MUST carry the SAME description text. Pre-build gate `CM-HUMAN-READABLE-DESCRIPTION` (when implemented) walks every item in the DB and asserts the `description` field exists, is non-empty, and is $\geq$ `11.4.171` characters. Paired `$` `11.4.171` meta-test mutation strips a description $\rightarrow$ gate FAILs. Classification: universal (`$` `11.4.171`). Composes `$` `11.4.171` (item-status tracking), `$` `11.4.171` (item-type tracking), `$` `11.4.171` (column alignment), `$` `11.4.171` (summary clarity), `$` `11.4.171` (SQLite SSoT), `$` `11.4.171` (item integrity), `$` `11.4.171` (universal export). Propagation gate `CM-COVENANT-114-171-PROPAGATION` (literal `11.4.171`) + paired `$` `11.4.171` mutation.

`$` `11.4.171` – **Mandatory production-readiness planning with realistic timeline projection** (User mandate, `$` `11.4.171`). Every project under this Constitution MUST maintain a living production-readiness planning document that: (a) provides REALISTIC timeline projections for all product milestones (MVP, beta, production-ready) based on actual development velocity data (items completed per week, reopening rate, blocking dependencies); (b) identifies ALL danger zones, weaknesses, and risk areas with mitigation strategies; (c) accounts for HW delivery timelines, licensing delays, and external dependency risks; (d) defines the critical path and identifies which items can slip without affecting MVP; (e) includes workstation/build-capacity analysis and optimization recommendations; (f) is updated at least monthly or whenever the item count changes by $\geq$ `11.4.171` %; (g) is cross-referenced against the workable-items database for accuracy. The planning document MUST live at `docs/research/production_planning_<YYYYMMDD>/ANALYSIS.md` with

HTML+PDF exports per. § . . . Honest boundary
 (§ . . .): projections are estimates based on measured
 velocity – they guarantee the methodology is sound, NOT that
 specific dates will be met. Classification: universal
 (§ . . .). Composes § . . . (no-guessing –
 projections based on data, not wishful thinking),
 § . . . (full retest before tag), § . . .
 (iteration discipline), § . . . (SQLite SSoT),
 § . . . (four-layer verification). Propagation gate
 CM-COVENANT-114-172-PROPAGATION (literal 11.4.172) + paired
 § . . . mutation.

§ . . . – **Containerized + distributed build mandate**
 (User mandate, – –). EVERY build of EVERY
 component (source compile, artifact/package/installer/
 container-image production, codegen, asset render – for ANY
 language/platform: Go, Android/Gradle, desktop, web, native,
 firmware) MUST run INSIDE a specialized build container
 provisioned via the digital.vasic.containers submodule
 (§ . . .) – NEVER on the bare host. The build
 containers MUST be DISTRIBUTED to the designated remote build
 host(s) (e.g. thinker.local) via the SAME containers-
 submodule distribution mechanism the infra uses
 (§ . . . Distributor / remote compose over SSH,
 § . . . rootless), so the build EXECUTES on the
 remote build host (offloading the developer/main host); once
 the build completes the produced artifacts MUST be brought
 BACK to the originating main host (scp/rsync/volume copy) for
 use/flashing/distribution. Building outside a container, or
 on the bare host, is FORBIDDEN – a release blocker (the
 “works on my machine” / unreproducible-build class
 § . . . exists to prevent). The build-host target +
 build-container definitions are config-injected
 (§ . . . , never hardcoded in the submodule); a missing
 build-container capability is added by EXTENDING the
 containers submodule upstream (§ . . .), never by an
 ad-hoc host build. Honest boundary (§ . . .): the

containerized+distributed build guarantees reproducibility + host-isolation + capacity offload – it does NOT replace the \$ 11.4.10. . . full-suite retest, \$ 11.4.11. . . four-layer artifact→runtime verification, or \$ 11.4.12. . . installable-asset evidence (those still run against the brought-back artifact). Classification: universal (\$ 11.4.24. . .). Composes \$ 11.4.82. . . (containers submodule – sole orchestration layer), \$ 11.4.83. . . (rootless runtime), \$ 11.4.84. . . (extend-don't-reimplement), \$ 11.4.85. . . (config injection / decoupling), \$ 11.4.86. . . (build-resource stats), \$ 11.4.87. . . (iteration speedup – persistent caches in the container), \$ 11.4.88. . . (no-commit-while-build-writes-artifacts), \$ 11.4.89. . . (installable-asset evidence on the brought-back artifact), \$ 11.4.90. . . (artifact→runtime verification), \$ 11.4.91. . . (host memory ceiling – offloading the build preserves the main host). Propagation gate CM-COVENANT-114-173-PROPAGATION (literal 11.4.173) + recommended gate CM-CONTAINERIZED-DISTRIBUTED-BUILD (every build runs via the containers submodule on the remote build host; a bare-host build is detected + FAILs) + paired \$ 11.4.92. . . mutation.

Canonical authority: constitution submodule Constitution.md \$ 11.4.174. . . . Non-compliance is a release blocker. No escape hatch – no 2024 10 29 --build-on-host , --skip-container-build , --local-build-ok , --no-distributed-build , --bare-host-build flag.

\$ 11.4.93. . . – Shared-host process-ownership verification mandate (User mandate, 11.4.94. . .). On any shared host where other projects/users/agents may run concurrently, every inspection of or action on a process, build, daemon, port, lock, or system-state signal MUST first positively VERIFY the target is OURS before diagnosing or acting. Attribute a process to us ONLY by a strong discriminator – cwd inside this project's tree, argv naming our scripts/artifacts/repo path, a PID we launched + recorded (\$ 11.4.95. . .), or a lock/port path under our tree – NEVER a loose pgrep java/gradle/node name match (it matches every

project on the host). NEVER kill/signal/restart a process not positively ours (other projects' gradle/java/podman/zig are off-limits — operator-workload-safety, \$ / \$. .); the \$. daemon-kill remediation MUST be scoped to OUR daemons. When our work waits on host contention from another project's process, surface it (\$. . /\$. .) — block-don't-break. Every pgrep/ps filter MUST exclude self-matches + other-project matches. Forensic anchor (FACT - -): a %CPU java+gradlew on the shared host was a separate catalogizer project's gradle test (cwd=Projects/catalogizer), nearly mis-attributed to our build. Composes \$. . / \$. . /\$. . /\$. . /\$. . /\$. . . Classification: universal (\$. . .). Propagation gate CM-COVENANT-114-174-PROPAGATION (literal 11.4.174) + recommended gate CM-PROCESS-OWNERSHIP-VERIFIED + paired \$. mutation. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch — no --assume-process-ours , --loose-pgrep-ok , --kill-any-matching-process , --skip-ownership-check flag.

\$. . — Conflict-free multi-track work-division + exactly-once claim registry + capability-aware deadlock-proof device-lock (User mandate, - -). Two mandatory multi-track arbitration layers, conflict-free by construction: (A) exactly-once work-item/logical-group claim registry (extends \$. . /\$. . ; disjoint file-scope \$. . L + worktree/CoW \$. . ; no codebase loss \$. /\$. . /\$. . /\$. .); (B) capability-aware deadlock-proof device-lock (\$. . single-owner; all-or-nothing + non-blocking + TTL-reap breaks all Coffman conditions; append-only JSONL + atomic snapshot \$. .); (C) universal/decoupled (\$. . / \$. . /\$. .) + evidence-based resource-tuning (\$. . /\$. . , never guessed).

Classification: universal (§ . .). Gates CM-WORK-DIVISION-EXCLUSIVE-CLAIM + CM-DEVICE-LOCK-DEADLOCK-FREE + CM-COVENANT-^{116 11 4 113} - -PROPAGATION + paired § . mutation.

Composes §^{12 6 . 2 .} § . . § . .

§^{11 4 176 . .} § . . § . . § . .

§ . . § . . § . .

§^{12 11 . .} § . . § . .

§ . . § . . §^{2026 07 03} . § .

§ . . .

§ . - Maximal dynamic resource utilization for containerized builds (User mandate, - -). The containerized build path (its OWN cgroup + own MemoryMax → OOM-kills ITSELF, never user.slice) MUST use the maximal SAFE fraction of host capacity, computed DYNAMICALLY per host: auto-detect nproc/MemTotal/RLIMIT_NPROC (never hardcode, § . .), reserve explicit host headroom (cores+RAM), scale -j against BOTH the memory model AND the process rlimit (privileged nproc-raise for full -j; EAGAIN-safe -j otherwise), never break/bottleneck/wedge. NO fixed low -j for the containerized path. The § . . % ceiling + bare-metal -j cap REMAIN, SCOPED to user.slice-resident work - §^{12 8 12 10 11 4 6} does NOT weaken § . . , it scopes it.

Classification: universal (§ . .). Gate CM-MAXRES-DYNAMIC-BUILD + CM-COVENANT- - -PROPAGATION + paired

§^{11 4 177} . mutation. Composes § . . § . . § . .

§ . . § . . § . . § . .

§^{2026 07 04} . . § . . § . . § . .

§ . . - Developer-tooling project-decoupling + invocation-directory operation (User mandate, - -). Compact summary: no project-specific script / hook / alias / binary may be wired (copied / symlinked / PATH-exported / aliased) into a global or shared developer-tooling PATH; shared tooling MUST be project-agnostic and operate on the INVOCATION directory (cwd / explicit path / config arg), NEVER a hardcoded project path;

a project-specific helper lives in + runs from its own repo, a genuinely-reusable helper is promoted to the shared toolkit ONLY after being stripped of every project-specific assumption (paths, device serials, package names, region endpoints) + taking its target from the invocation context; a project script reachable on the global/shared PATH is a decoupling violation of equal severity to importing project-specific code into a shared submodule (\$. .); forensic FACT - - - a project cwd-hook symlinked into the global Claude-Toolkit PATH re-coupled every alias to one hardcoded checkout; honest boundary \$. . - decoupling guarantees project-agnostic, not correct (still needs the tool's own tests). Classification: universal (\$. .). Composes \$. . . / . . / . . / . . . Propagation gate CM-COVENANT-114-177-PROPAGATION (literal 11.4.177) + recommended gate CM-TOOLING-PROJECT-DECOUPLED + paired \$. . mutation. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch - no --global-symlink-ok , --hardcode-project-path , --wire-into-shared-path flag.

\$. . - Track-qualified identity for parallel work streams (session-name collision ban) (User mandate, - -). Compact summary: parallel work streams sharing hardware / a git backing store / a tmux-session namespace / a lock namespace / any single-owner resource MUST be addressed by a TRACK-QUALIFIED identity (<project>__<track>__<role>), NEVER a bare directory basename; multiple checkouts/clones/worktrees sharing basename atmosphere collapse into one session/lock/log key + cross-wire tmux targets, lock files, device claims, log dirs (observed collision - . .); every registry row / session name / lock path / log dir / device claim for a ≥ -concurrent-claimant resource MUST carry the track-qualified key, a bare-basename identity for such a resource is a violation; honest boundary \$. . - a unique

identity prevents cross-wiring, it does not itself arbitrate ownership (that is \$. . . + \$. . . (the multi-track work-division + exactly-once-claim + device-lock arbitration anchor)). Classification: universal

(\$. . .). Composes

\$. . . /. /. . /. + \$. . . (the multi-track work-division + exactly-once-claim + device-lock arbitration anchor). Propagation gate CM-COVENANT-114-178-PROPAGATION (literal 11.4.178) + recommended gate CM-TRACK-QUALIFIED-IDENTITY + paired \$. . . mutation. **Canonical authority:** constitution submodule Constitution.md

\$ Non-compliance is a release blocker. No escape hatch – no --bare-basename-ok , --skip-track-qualifier , --shared-session-name flag.

\$. . . – Corruption-isolated parallel git streams (own-.git independent clones, not shared-common-dir worktrees) (User mandate, – –). Compact summary: when corruption-isolation is a requirement, parallel git work streams MUST each be an isolated repo with its OWN .git (own object store, own index, own lock namespace), NOT .git worktree checkouts sharing one common .git ; a shared common-dir is a single point of failure – one stale index.lock / .commit_all.lock / .push_all.lock freezes EVERY stream at once (observed -h all-track freeze – –) + one corrupted object store loses every stream; each stream's git rev-parse --git-common-dir MUST resolve INSIDE its own tree, a common-dir resolving to a shared parent is a violation; where CoW/reflink disk-feasibility is the constraint use the \$. . . reflink-clone-with-own-.git path (btrfs/XFS/ZFS/APFS reflink shares extents on disk while keeping a per-stream .git , so isolation + disk-feasibility both hold); honest boundary

\$. . . – own-.git contains lock/corruption blast radius, it does NOT remove per-repo stale-lock reaping (\$. . .) nor ff-only no-force integration (\$. . .). Classification: universal

(`$ . .`). Composes `$ . . /.` `/.` `/`
`$ .` + `$ . .` (the multi-track work-division +
exactly-once-claim + device-lock arbitration anchor).
Propagation gate `CM-COVENANT-114-179-PROPAGATION` (literal
`11.4.179`) + recommended gate `CM-GIT-STREAMS-OWN-COMMON-DIR` +
paired `$ .` mutation. **Canonical authority:** constitution
submodule `Constitution.md` `$ . .`. Non-compliance is
a release blocker. No escape hatch – no `--shared-common-git`,
`--worktree-when-isolation-required`, `--skip-git-isolation` flag.

`$ . .` – Commit/push (single-writer) wrappers **MUST**
auto-reap provably-stale locks before acquiring (User
mandate, – –). Compact summary: every
commit / push / single-writer wrapper **MUST**, before acquiring
its own lock, auto-reap any PROVABLY-stale git lock – one
whose recorded holder PID is DEAD (`kill -0` fails), OR (no PID
recorded) older than a defined threshold AND with no live
wrapper/git process holding that repo; the reap covers the
wrapper's own lock plus the git-internal existence-based
locks (`index.lock`, `HEAD.lock`, `refs/**/*.lock`) git.does NOT
auto-release on holder death; it **MUST NEVER** remove a lock
whose holder is ALIVE (removing a live-held lock corrupts a
concurrent writer – `$ .`); each reap decision (REAPED /
KEPT + reason) is logged as captured evidence (`$ . .` –
liveness PROVEN via `kill -0`, never assumed); a wrapper that
blocks indefinitely on a dead-holder lock (observed `-h`
freeze `11 4 14 13 - 0 9 2 11 4`) is a violation; honest boundary
`11 4 179 . .` – reaping clears dead-holder deadlock, it does
NOT substitute for the own-`.git` isolation of
`$ . .`. Classification: universal (`$ . .`).
Composes `$ . . /.` `/.` `/``$ . /``$. . /`
`$ . .`. Propagation gate `CM-COVENANT-114-180-`
`PROPAGATION` (literal `11.4.180`) + recommended gate `CM-WRAPPER-`
`STALE-LOCK-AUTO-REAP` + paired `$ .` mutation. **Canonical**
authority: constitution submodule `Constitution.md`

\$. . . Non-compliance is a release blocker. No escape hatch – no `--skip-stale-lock-reap`, `--force-remove-live-lock`, `--block-on-dead-holder` flag.